Cattle IoT Master Document (Godhan App Integration)

This document is a **living master file** for the cattle IoT system (ESP32 collars + Godhan app integration). It contains both high-level functional design and technical development details.

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1. Hardware Design

- ESP32-based collar with:
- Accelerometer (IMU) for activity.
- Temperature sensor (DS18B20 / IR / rumen bolus).
- Microphone (MEMS + FFT) for rumination.
- Battery pack (Li-ion/LiPo \~3400mAh).
- Features:
- Wi-Fi connectivity (farm Wi-Fi assumed).
- Data sync every 5 minutes.
- OTA firmware updates.
- Fail-safe local storage (SPIFFS).
- Low-battery alerts.

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2. Firmware Features

- Sample IMU, temperature, rumination mic.
- Aggregate + send via MQTT/HTTP.
- Deep-sleep between transmissions.
- Local buffer when offline.
- OTA firmware via HTTP/MQTT trigger.
- Configurable via BLE captive portal or app pairing.

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3. Data Flow Architecture

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ESP32 Collar → MQTT Broker → Backend → Godhan App
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- ESP32 collects → publishes via MQTT.
- Backend ingests → DB storage + analytics → alerts.
- App displays cattle data, alerts, and triggers OTA.

Sequence Flows

- **Device Assignment:** App pairs via BLE → sends Wi-Fi + cattle_id → ESP32 registers with backend.
- **Data Sync:** ESP32 publishes MQTT payload → backend stores + processes → app fetches via API.
- **OTA Update:** Admin uploads firmware → backend notifies ESP32 → ESP32 downloads + flashes.

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4. Database Schema (Concise)

- **cattle**: profiles, breed, insemination, pregnancy, expected/actual calving.
- **devices**: collars, firmware, status, battery.
- **sensor_data**: time-series telemetry.
- **alerts**: estrus, calving, fever, battery.
- **firmware**: available versions, URLs.
- **farms**: farm info, Wi-Fi.
- **users**: farmers, vets, admins.

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5. Technical Model Schemas (Mongoose)

- **Cattle**: cattle_id, farm_id, breed, lactation_stage, insemination_date, pregnancy_confirmed, expected_calving_date, actual_calving_date.
- **Device**: device_id, cattle_id, fw_version, last_seen, battery, status.
- **SensorData**: cattle_id, timestamp, temp, accX, accY, accZ, rumination_events, battery.
- **Alert**: alert_id, cattle_id, type, severity, message, timestamp.
- **Firmware**: fw_version, url, notes, status.
- **Farm**: farm_id, owner_id, location, wifi_ssid.
- **User**: user_id, role, farms.

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6. API Endpoints (Concise)

- **Devices**: register, assign, get status.
- **Data**: post sensor data, get cattle data, get cattle alerts.
- **Firmware**: trigger update, get latest version.
- **Events**: insemination, pregnancy confirmation, calving log.

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7. Heat Detection Module

- **Signals:** activity ↑, rumination ↓, subtle temp changes.
- **Rules (MVP):** activity > baseline + 2σ AND rumination < 80% baseline → heat alert.
- **ML Upgrade:** RandomForest/XGBoost on window features (activity spikes, rumination drop, temp trend).
- **Output:** estrus probability (0–1), best insemination window = 6–24h after heat onset.

8. Calving Prediction Module

- **Step 1:** Expected calving date = insemination_date + \~283d.
- **Step 2:** Monitor signals (temp drop \~0.5°C, rumination drop >30%, activity restlessness).
- **Alerts:**
- 14d: start monitoring.
- 7d: show approaching.
- 3d: prepare pen.
- Imminent: strong signal → calving in next 12–24h.
- **DB Additions:** insemination_date, pregnancy_confirmed, expected_calving_date, actual_calving_date, calving_prediction_history.

9. Dashboard (App/Grafana)

- **Farm Overview:** herd health, estrus, calving alerts, device status.
- **Cattle Detail:** 14/7/3-day trend graphs (temp, rumination, activity).
- **Heat Detection:** candidate cattle + probability.
- **Calving Prediction:** due dates, imminent calving alerts.
- **Device Maintenance:** firmware status, offline list.

10. Security Model

- Device → TLS + HMAC payload signing.
- App ↔ Backend → JWT auth + HTTPS.
- OTA → signed firmware, validated by device.
- Data privacy → encryption at rest + role-based access.

11. Roadmap

1. Pilot (10–20 cattle, Wi-Fi collars).
2. Heat detection rules → app alerts.
3. Add insemination/pregnancy tracking.
4. Calving prediction module (14/7/3/1 day alerts).
5. Scale with ML models, OTA updates, dashboards.

12. Open Items

- Edge vs cloud analytics.
- Power strategy (solar collars?).

• Feedback loop in app (farmer confirms heat/calving → improves models).
• Integration with vet ERP + buyer dashboards.
